COP3530 Data Structures and Algorithms
Project 2 Proposal

Goobers | Saving Tampa Bay | Group 59 | Aditya Nair, Suhas Gaddam, Vishnu Araveeti

I. Problem

During hurricanes and severe flooding, road closures and hazardous conditions make traditional shortest-path navigation unreliable for planning. In coastal cities, the shortest route to a highway or exfil location can very easily become submerged based on elevation. This project aims to develop a flood-aware evacuation routing system using OSM road network data to identify optimal routes over a flooded tampa bay road graph network. By simulating flood conditions and closures, we can evaluation how hazards affect path finding with Dijkstra and A* pathfinding algorithms.

II. Motivation

Florida frequently experiences hurricanes and flooding, making evacuation planning an important issue in the real world. As all of our members are Tampa residents and have experienced such hurricanes, we take a personal stake in this issue.

III. Features

The project will allow users to select an evacuation starting point from several Poi's across high traffic areas in Tampa, with the designated evacuation channels as targets. Users will be able to set the flood severity to modify road access and travel costs. Using Djikstra and A* we will calculate optimal routes, and export them on a map highlighting the recommended route and affected roads. Additionally a performance comparison will display execution time, nodes explored, and route cost to evaluate relative efficiency.

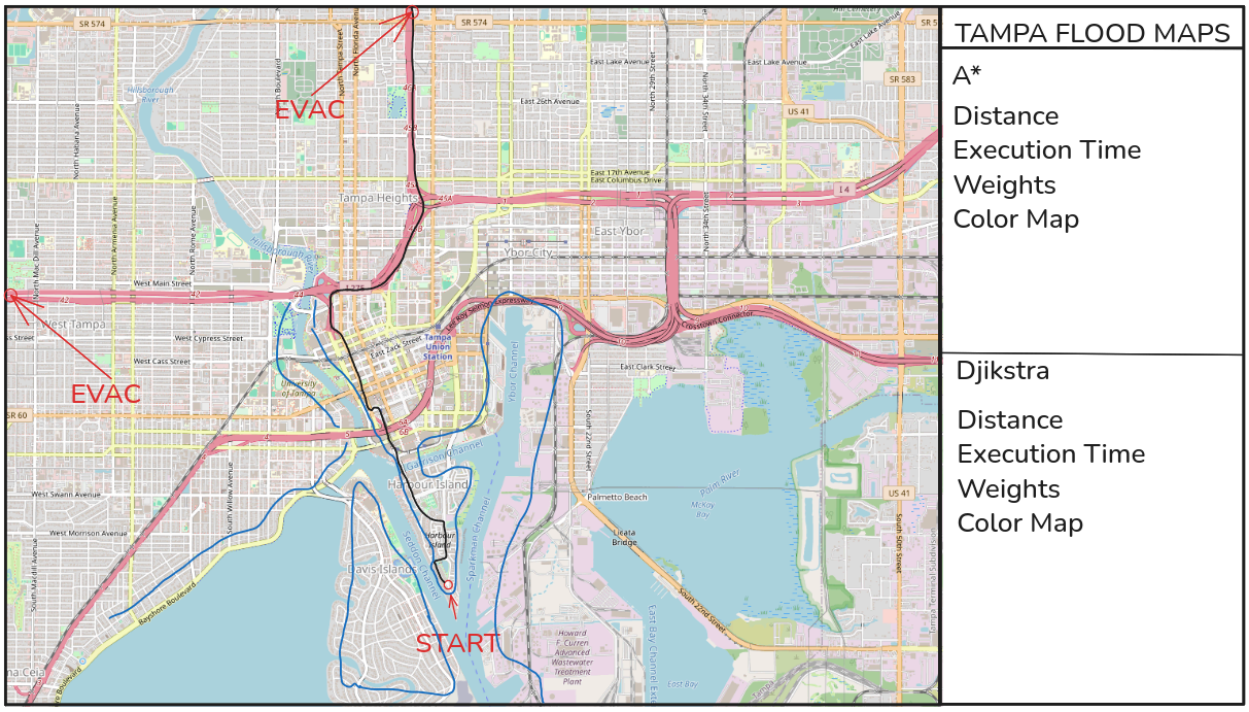
IV. Data

The project will work on public road network data from OpenStreetMap (OSM) extracted via the python OSMnx library covering Tampa Bay with a focus on high foot traffic down town areas, sized to fit 100,000 street segments. We'll represent it as a directed graph, with intersections as nodes containing Id's, latitude, longitude, and weighted by length. Additional simulated flood-risk values will be assigned to each segment based on distance to water and elevation, with each value effecting the algorithm's weights.

V. Tools

The project will primarily use C++ to implement Dijikstra's and A* pathfinding algorithms from scratch, along with graph construction, flood simulation, and performance benchmarking. Python will be used for data preprocessing and visualization. The OSMnx library will retrieve and process real road network from OSM, which will be exported into CSV files for use in C++. Matplotlib and OSMnx will generate PNG maps displaying the evacuation routes and simulated flooded roads. We will also use the C++ Library to provide basic data structures and file handling.

VI. Visuals



**Rendered PNG product, actual tool will be a CLI binary outputting the product for viewing*

VII. Strategy

We plan to implement Dijikstra's algorithm and A* Search in C++ to compare their efficiency in finding evacuation routes. The road network will be represented as a weighted, directed, graph using an adjacency list, where vertices represent road intersections and edges represent roads with weights based on distance and simulated flood severity. We will use a min-priority queue to efficiently select the next node to explore, and A* will use a straight-line distance algorithm to guide its search. We will compare both algorithms based on execution time, nodes explored, and the total route cost and safety.

VIII. Distribution of Responsibility and Roles

Aditya will focus on obtaining and processing the OSM data and building the graph representation. Suhas will focus on implementing the two algorithms, Dijikstra's and A*, while Vishnu will handle the map visualizations and result generation. We will all work together implementing the additional features, testing, performance comparisons, and the final presentation.

IX. References

OSMnx 2.1.1 documentation. (n.d.). https://osmnx.readthedocs.io/en/stable/